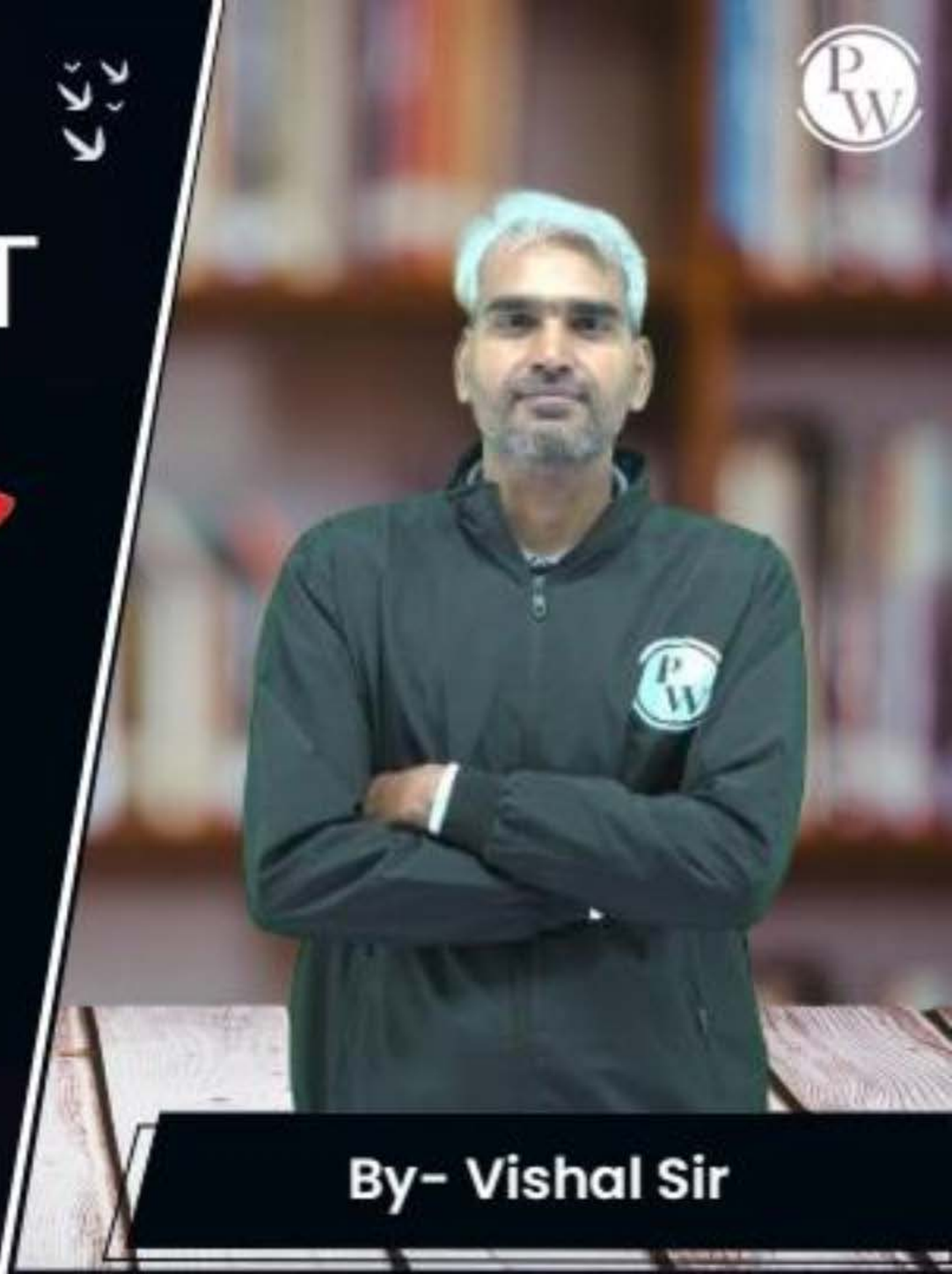
Computer Science & IT



Compiler Design

Syntax Analysis (Parser)

Lecture No. 01



By- Vishal Sir

Recap of Previous Lecture



Topic Tokens in C

Topic Questions

Slide

Topics to be Covered



Topic Quick Revision

Topic Introduction to Syntax Analyzer

Slide



Topic : Quick Revision

#Q. Consider the following program segment.

```
int max(i,j)
{
    int i,j;
    /*function returns
                maximum of i and j*/
    return i>j?i:j;
}
```

A. Lexical Error

B. No Lexical Error and Number of tokens created = 22

C. No Lexical Error and Number of tokens created = 23

Slide

// Every thing is comment

/* Kaha gaya bhai */

// Kaha gaya bhai

// Kaha gaya bhai

identifier

every thing after // is treated as comment

```
int x, y //x & y are integer type variable//;
```

```
int x, y /*x & y are int type variable*/;
```

```
//*- - - -
```



Topic : Quick Revision

#Q. Consider the following program segment.

```
int a, ab ;
```

A. Lexical Error

B. No Lexical Error and Number of tokens created = 5

C. No Lexical Error and Number of tokens created = 6

D. None of the above

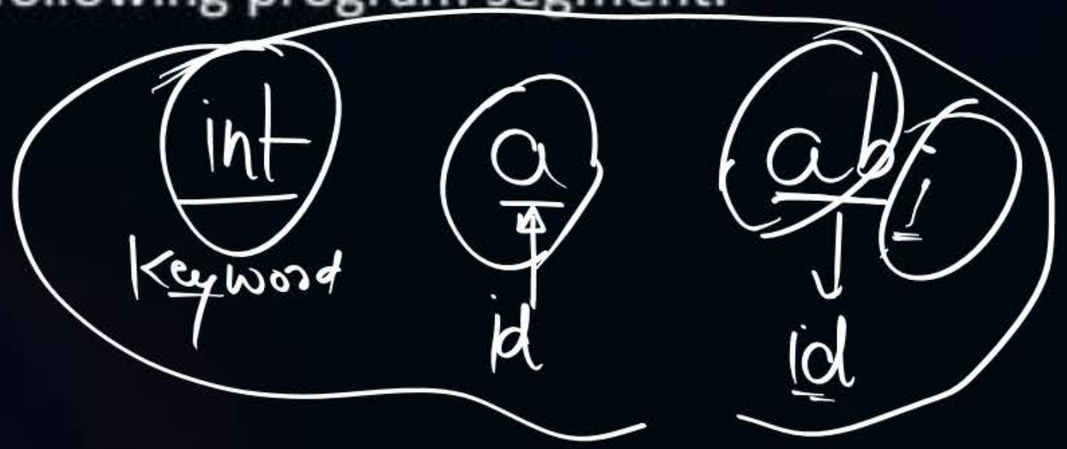
Slide

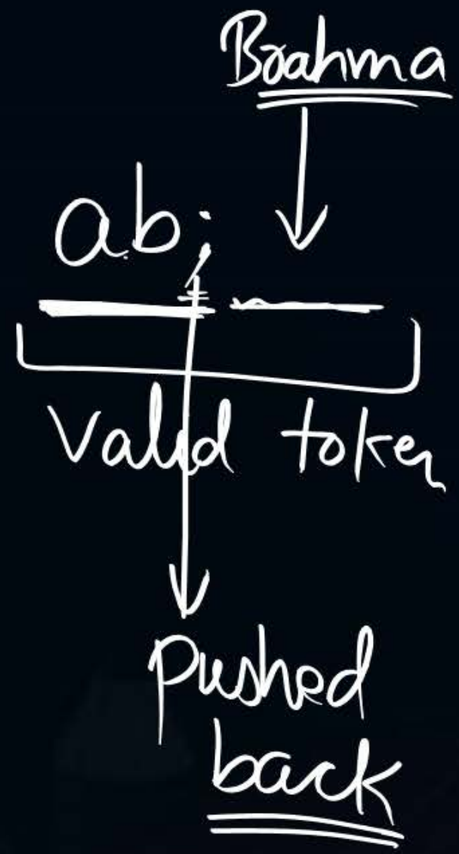


Topic : Quick Revision

#Q. Consider the following program segment.

```
int a ab ;
```





A. Lexical Error

B. No Lexical Error and Number of tokens created = 4

C. No Lexical Error and Number of tokens created = 5

D. None of the above

Slide



Topic : Quick Revision

#Q. Consider the following program segment.

```
int a, ab, ab1, 12, ab12cd ;
```

Meaningful sequence of characters defining a Constant

Constant

No lexical error

A. Lexical Error

B. No Lexical Error and Number of tokens created = 11

C. No Lexical Error and Number of tokens created = 13

D. None of the above

123 gate

Constant Space identifier

123gate → not a numeric Constant

digit, ∴ not an identifier

lexical error

Slide



Topic : Quick Revision

#Q. Consider the following program segment.

```
int a, ab, abc ab12, _ab, a_b a123, 23, 23a;
```

A. Lexical Error

B. No Lexical Error and Number of tokens created = 15

C. No Lexical Error and Number of tokens created = 16

D. None of the above = 18

Slide



Topic : Quick Revision

#Q. Consider the following program segment.

```
int a, ab , 23a;
```

A. Lexical Error

B. No Lexical Error and Number of tokens created = 8

C. No Lexical Error and Number of tokens created = 7

D. None of the above



Topic : Quick Revision

"single Constant"

#Q. Consider the following program segment.

```
printf ( "Hello students ans = %d", i );
```

Single token = String Constant

A. Lexical Error

B. No Lexical Error and Number of tokens created = 14

C. No Lexical Error and Number of tokens created = 7

D. None of the above



Topic : Quick Revision

#Q. Consider the following program segment.

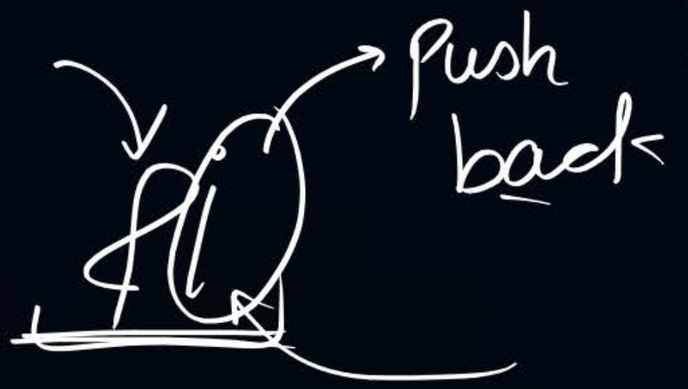
```
printf("i = %d , &i  = %x", i, &i);
```

A. Lexical Error

B. No Lexical Error and Number of tokens created = 10

C. No Lexical Error and Number of tokens created = 9

D. None of the above



1. Key
2. Id
3. Op
4. Const
5. Sp. Sym
6. String

Slide



Topic : Quick Revision

int &i;

i ; → Push token

#Q. Consider the following program segment.

```
main()
{
    fro(i=0;i<=10;i++)
}
```

will be treated as an identifier by L.A

foo(_ , _ , _)

not syntactically correct w.r.t grammar of C language

∴ Syntax Error.

A. Lexical Error

B. No Lexical Error and Number of tokens created = 18

C. No Lexical Error and Number of tokens created = 19

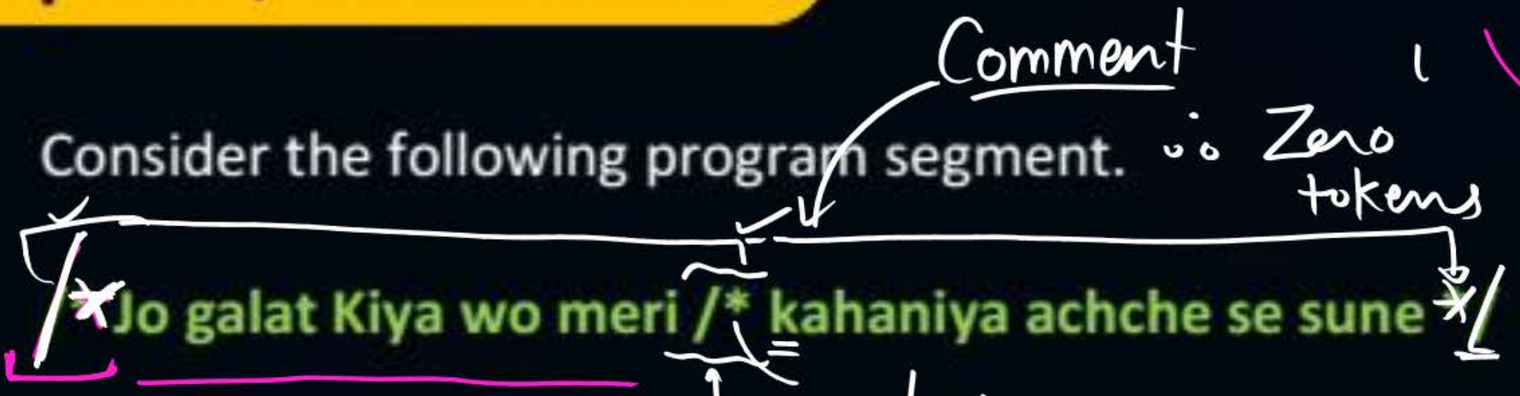
D. None of the above

Slide



Topic : Quick Revision

#Q. Consider the following program segment.



∴ Zero tokens

does not give importance to any seq. of character until */ is found



What if it never appears? then error

A. Lexical Error

B. No Lexical Error and Number of tokens created = 7

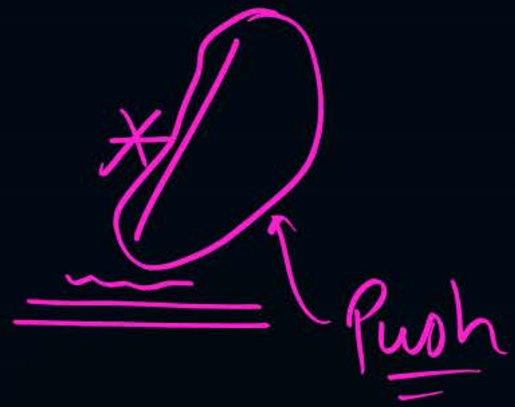
C. No Lexical Error and Number of tokens created = 0

D. No Lexical Error and Number of tokens created = 1

Slide



Topic : Quick Revision



#Q. Consider the following program segment.

```
/* Ye bhi batao */ to Mane warna kya fayda */
```

Comment

A. Lexical Error

B. No Lexical Error and Number of tokens created = 7

C. No Lexical Error and Number of tokens created = 0

D. No Lexical Error and Number of tokens created = 1



Topic : Quick Revision

#Q. Consider the following program segment.




```
/* And cds_7 do it*/ /**/ 1a */
```

Single Comment

lexical Error

A. Lexical Error

B. No Lexical Error and Number of tokens created = 3

C. No Lexical Error and Number of tokens created = 0

D. No Lexical Error and Number of tokens created = 4

Slide

Note :- * Lexical Analyzer can be implemented using tools such as lex.

OR

Lexical analyzer can be hardcoded by implementing finite automatas.

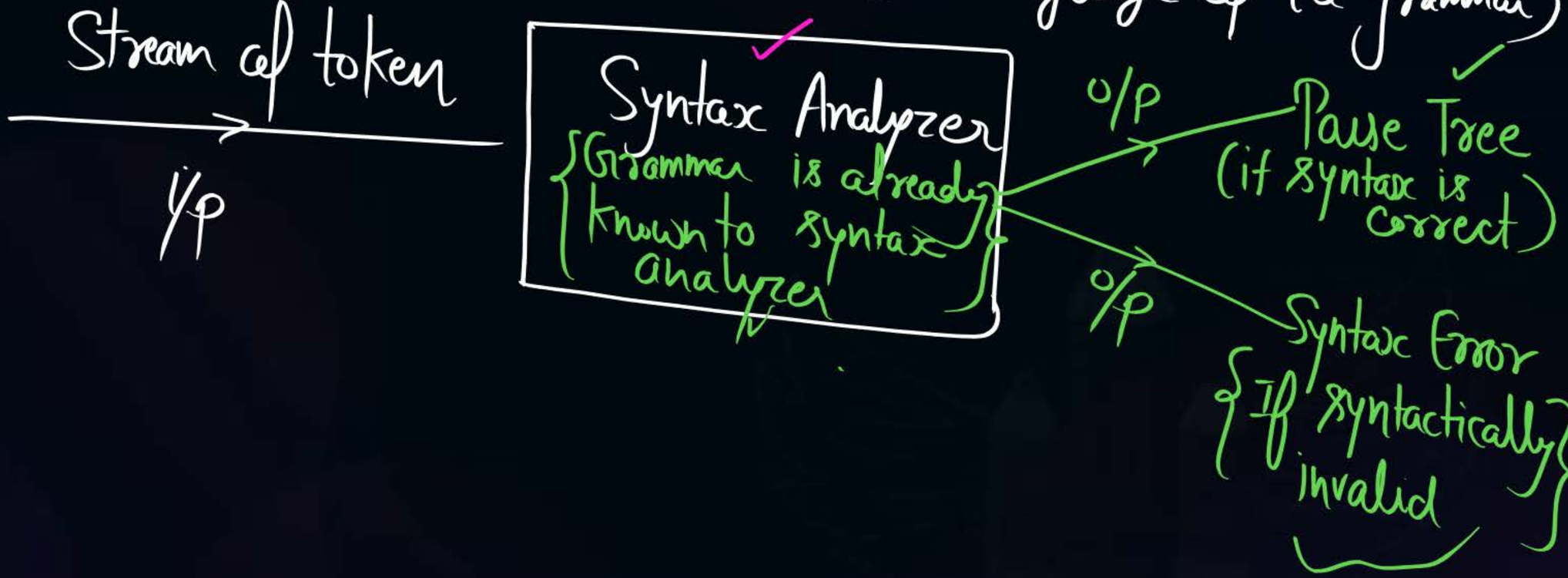
* Regular expressions can be used to represent the patterns of different types of tokens



Topic : Syntax Analyzer (Parser)

functionality :-

To Check whether the statement is syntectically Correct or not based on the language of the grammar



Slide



Topic : Syntax Analyzer (Parser)

Syntax analyzer is a program that takes sequence of tokens as an input and tries to verify whether the sequence of tokens can be derived using the grammar known to the syntax analyzer or not.

If sequence of token (input string) is derivable from the grammar i.e., input string is member of language generated by grammar, then syntax analyzer produces the parse tree as an output otherwise it returns syntax error to the error handler.



2 mins Summary

Topic Lexical Errors

Topic Basic understanding of syntax analyzer, grammar and derivation tree.

THANK - YOU